Metric Hilbert Factorization Does Not Imply Multilinear Hilbert Factorization

Automatic mathematics run `fa_banach_001`

9 August 2026

Status. Candidate counterexample, likely valid, for human review. The construction is over the real scalars. It answers both Questions 1 and 2 in Section 2.2 of Fernández-Unzueta–García-Hernández negatively.

1 Source question

The source is M. Fernández-Unzueta and S. García-Hernández, *Multilinear Operators Factoring through Hilbert Spaces*, arXiv:1805.09748, later published in *Banach Journal of Mathematical Analysis* 13 (2019), 234–254 [1].

For a multilinear operator $T : X_1 \times \cdots \times X_n \to Y$, the paper writes $f_T : \Sigma^{\pi}_{X_1,\ldots,X_n} \to Y$ for the restriction of the linearization of T to the Segre cone. It proves

$$\gamma_2^{\mathrm{Lip}}(f_T) \le \Gamma(T),$$

where $\Gamma(T)$ is the multilinear Hilbert-factorization norm and γ_2^{Lip} is the metric factorization norm through a subset of a Hilbert space. The authors ask:

Question 1. Is the map $T \mapsto f_T$ an isometry?

Question 2. If f_T factors through a Hilbert space in the metric sense, must T factor through a Hilbert space in the multilinear sense?

The statement occurs across printed pages 7–8. The two source crops are included below.

where μ is a probability measure on $(B_{\mathcal{L}(X_1,\ldots,X_n)^*}, w^*)$, $i : X_1 \times \cdots \times X_n \to C(B_{\mathcal{L}(X_1,\ldots,X_n)^*})$ acts by evaluation, $j_2 : C(B_{\mathcal{L}(X_1,\ldots,X_n)^*}) \to L_2(\mu)$ is the canonical inclusion and u is a Lipschitz function such that $\pi_2^{Lip}(T) = Lip(u)$. Hence $T \in \Gamma(X_1,\ldots,X_n;Y)$ and $\Gamma(T) \leq \pi_2^{Lip}(T)$.

Actually, we have proved that

$$\|Id : \Pi_2^{Lip}(X_1,\ldots,X_n;Y) \to \Gamma(X_1,\ldots,X_n;Y)\| \leq 1$$

holds for all Banach spaces $X_1,\ldots,X_n$ and Y.

2.2. **Relation with the Metric Case.** Now, we turn our attention to Lipschitz mappings between metric spaces. Recall from [4] that a Lipschitz function between metric spaces $f : X \to Y$ factors through a subset of a Hilbert space if there exist a Hilbert space H and a subset Z of H (actually we may take $Z = f(X)$) and two Lipschitz functions $A : X \to Z$, $B : Z \to Y$ such that $f = BA$. In this case $\gamma_2^{Lip}(f) = \inf Lip(A)\, Lip(B)$, where the infimum is taken over all possible factorizations of f as before.

It is clear from Diagram (4) that if T is an element in $\Gamma(X_1,\ldots,X_n;Y)$, then its associated Σ-operator $f_T : \Sigma^\pi_{X_1\ldots X_n} \to Y$ is a Lipschitz function that can be factored through a subset of a Hilbert space in the sense of [4]. Moreover, we have

$$\gamma_2^{Lip}(f_T) \leq \Gamma(T).$$

In other words, every multilinear operator T in $\Gamma(X_1,\ldots,X_n;Y)$ gives rise to a Lipschitz function f_T in $\Gamma_2^{Lip}(\Sigma^\pi_{X_1\ldots X_n};Y)$. That is, the operator

$$\begin{array}{rcl} \Gamma(X_1,\ldots,X_n;Y) & \to & \Gamma_2^{Lip}(\Sigma^\pi_{X_1\ldots X_n};Y) \\ T & \mapsto & f_T \end{array} \tag{5}$$

is bounded and has norm ≤ 1.

We do not know if the metric approach of [4] restrict well to the setting of multilinear operators we are proposing. Specifically, we have two questions:

Question 1: Is the map defined in (5) an isometry?

Question 2: We do not know if T factors through a Hilbert space whenever f_T does in the metric sense, that is, Does f_T in $\Gamma_2^{Lip}(X;Y)$ imply T in $\Gamma(X_1,\ldots,X_n;Y)$?

3. Kwapień Type Characterization

2 Definitions used in the construction

All spaces below are real. If $S : X_1 \times \cdots \times X_n \to Y$ is multilinear, then $S \in \Gamma(X_1,\ldots,X_n;Y)$ means that

$$S = B \circ A,$$

where A is a bounded multilinear map into a Hilbert space H, and B is Lipschitz on $A(X_1 \times \cdots \times X_n)$. The norm $\Gamma(S)$ is the infimum of $\|A\|\,\mathrm{Lip}(B)$.

For a Lipschitz map $f : M \to Y$, $\gamma_2^{\mathrm{Lip}}(f)$ is the infimum of $\mathrm{Lip}(a)\,\mathrm{Lip}(b)$ over factorizations $f = b \circ a$ with a mapping M into a subset of a Hilbert space and b Lipschitz there.

Write $E_m = \ell_1^m$ and identify

$$E_m \widehat{\otimes}_\pi E_m = \ell_1^{m\times m}$$

by the usual matrix coordinates. Let

$$J_m : E_m \times E_m \longrightarrow \ell_1^{m\times m}, \qquad J_m(x,y) = x\otimes y.$$

Its associated Segre operator is the identity map on the rank-one cone $\Sigma_m = \{x\otimes y : x,y\in E_m\}$, equipped with the ambient entrywise ℓ_1 metric.

3 Idea of the proof

There are two different geometries in the source comparison. A multilinear factorization of J_m must encode every sign tensor $\varepsilon\otimes\delta$ by a *bilinear* Hilbert-valued map. A Rademacher average then forces distortion at least m, and the elementary coordinate embedding shows that the exact value is $\Gamma(J_m) = m$.

The metric factor may encode the rank-one cone nonlinearly. A nonzero rank-one matrix has the form $r(u\otimes v)$, where r is its ℓ_1 norm and u, v are on the ℓ_1 unit sphere, uniquely up to the simultaneous sign change $(u,v)\mapsto(-u,-v)$. The quadratic Hilbert feature

$$(u,v) \longmapsto \frac{(u,v)\otimes(u,v)}{\|(u,v)\|_2}$$

removes exactly that sign ambiguity. Its distortion on the direction space is $O(\sqrt{m})$; adjoining the radial coordinate preserves this order on the whole cone. Hence $\gamma_2^{\mathrm{Lip}}(f_{J_m}) = O(\sqrt{m})$, strictly smaller than $\Gamma(J_m) = m$ for large m.

Finally, put increasingly large finite-dimensional examples on diagonal blocks and weight the mth block by the reciprocal of its metric bound. The Segre map then has a metric Hilbert factorization with constant 1, whereas the multilinear Γ-norms of the block restrictions tend to infinity. This produces one bounded bilinear operator outside Γ and answers Question 2.

4 A Hilbert embedding of the rank-one cone

Let

$$\mathcal{R}_m = \{u\otimes v : \|u\|_1 = \|v\|_1 = 1\} \subset \Sigma_m.$$

For $a = u\otimes v$ and $b = u'\otimes v'$ in $\mathcal{R}_m$, put

$$\begin{aligned}\rho(a,b) &= \|a-b\|_1,\\ \delta(a,b) &= \min_{\sigma\in\{-1,1\}}\big(\|u-\sigma u'\|_1 + \|v-\sigma v'\|_1\big).\end{aligned}$$

Lemma 1 (Stability of normalized rank-one factors)**.** *For all $a, b\in\mathcal{R}_m$,*

$$\rho(a,b) \le \delta(a,b) \le 8\rho(a,b).$$

Proof. For either sign σ,

$$u\otimes v - u'\otimes v' = (u-\sigma u')\otimes v + \sigma u'\otimes(v-\sigma v'),$$

so $\rho\le\delta$.

Suppose first that $\rho < 1/2$. Choose $f\in B_{\ell_\infty^m}$ with $f(u) = 1$. Contracting the first tensor coordinate gives

$$\|v - f(u')v'\|_1 \le \rho.$$

Thus $|f(u')| \geq 1-\rho > 1/2$. With $\sigma = \operatorname{sign} f(u')$,

$$\|v - \sigma v'\|_1 \leq 2\rho.$$

Choose $g \in B_{\ell_\infty^m}$ with $g(v) = 1$. Contracting the second coordinate gives $\|u - g(v')u'\|_1 \leq \rho$, while

$$|g(v') - \sigma| \leq \|v' - \sigma v\|_1 \leq 2\rho.$$

Consequently $\|u - \sigma u'\|_1 \leq 3\rho$, so in fact $\delta \leq 5\rho$. If $\rho \geq 1/2$, the trivial bound $\delta \leq 4$ gives $\delta \leq 8\rho$. □

For $a = u \otimes v \in \mathcal{R}_m$, set $w_a = (u, v) \in \mathbb{R}^{2m}$ and define

$$h_m(a) = \frac{w_a \otimes w_a}{\|w_a\|_2} \quad \text{in} \quad K_m := \mathbb{R}^{2m} \widehat{\otimes}_2 \mathbb{R}^{2m}.$$

The map is well-defined because the only ambiguity in normalized real rank-one factors is $(u, v) \mapsto (-u, -v)$. Also $\|h_m(a)\|_2 = \|w_a\|_2 \leq \sqrt{2}$.

Lemma 2 (Quadratic quotient feature)**.** *For $a, b \in \mathcal{R}_m$,*

$$\frac{\rho(a,b)}{\sqrt{2m}} \leq \|h_m(a) - h_m(b)\|_2 \leq 8\sqrt{2}\,\rho(a,b).$$

Proof. Let

$$d_2 = \min_{\sigma \in \{-1,1\}} \|w_a - \sigma w_b\|_2.$$

Write $\alpha = \|w_a\|_2$, $\beta = \|w_b\|_2$, and $c = |\langle w_a, w_b\rangle|/(\alpha\beta)$. Then

$$\begin{aligned} d_2^2 &= \alpha^2 + \beta^2 - 2\alpha\beta c, \\ \|h_m(a) - h_m(b)\|_2^2 &= \alpha^2 + \beta^2 - 2\alpha\beta c^2. \end{aligned}$$

Since $2\alpha\beta(c - c^2) \leq 2\alpha\beta(1-c) \leq d_2^2$,

$$d_2 \leq \|h_m(a) - h_m(b)\|_2 \leq \sqrt{2}\, d_2.$$

Moreover,

$$d_2 \leq \delta(a,b) \leq \sqrt{2m}\, d_2.$$

Combine these inequalities with Lemma 1. □

Every nonzero $p \in \Sigma_m$ has a unique representation $p = ra$, where $r = \|p\|_1 > 0$ and $a \in \mathcal{R}_m$. Define

$$F_m(0) = 0, \qquad F_m(ra) = \big(r, r h_m(a)\big) \in H_m := \mathbb{R} \oplus_2 K_m.$$

Proposition 3 (Metric factorization bound)**.** *The map F_m is bi-Lipschitz and*

$$\operatorname{Lip}(F_m) \leq 24\sqrt{2}, \qquad \operatorname{Lip}(F_m^{-1}) \leq 5\sqrt{m}.$$

Consequently

$$\gamma_2^{\mathrm{Lip}}(f_{J_m}) \leq 120\sqrt{2m}.$$

Proof. Take $p = ra$ and $q = sb$ and suppose $r \geq s \geq 0$. Put $d = \|p - q\|_1$ and $\rho = \rho(a, b)$ when $s > 0$ (when $s = 0$ the same estimates follow directly). The triangle inequality and its reverse give

$$r - s \leq d, \qquad s\rho \leq d + (r - s) \leq 2d,$$

hence

$$(r - s) + s\rho \leq 3d, \qquad d \leq (r - s) + s\rho. \tag{1}$$

Let $D = \|F_m(p) - F_m(q)\|_2$. By Lemma 2 and $\|h_m(a)\|_2 \leq \sqrt{2}$,

$$\begin{aligned} D &\leq (r - s) + \|rh_m(a) - sh_m(b)\|_2 \\ &\leq (1 + \sqrt{2})(r - s) + 8\sqrt{2}\, s\rho \leq 24\sqrt{2}\, d, \end{aligned}$$

using (1).

Conversely, $r - s \leq D$, and

$$s\|h_m(a) - h_m(b)\|_2 \leq \|rh_m(a) - sh_m(b)\|_2 + (r - s)\|h_m(a)\|_2 \leq (1 + \sqrt{2})D.$$

Lemma 2 and (1) now imply

$$d \leq (r - s) + s\rho \leq \left[1 + (1 + \sqrt{2})\sqrt{2m}\right] D \leq 5\sqrt{m}\, D.$$

Thus F_m has the stated constants. Since f_{J_m} is the identity on Σ_m, the factorization $f_{J_m} = F_m^{-1} \circ F_m$ proves the final bound. □

5 The exact multilinear norm

Proposition 4. *For every $m \geq 1$,*

$$\Gamma(J_m) = m.$$

Proof. For the upper bound, regard $A(x, y) = x \otimes y$ as taking values in the Hilbert space $\ell_2^{m\times m}$. Then $\|A\| = 1$. On the rank-one cone, let B be the identity into $\ell_1^{m\times m}$. Since $\|z\|_1 \leq m\|z\|_2$ for every $z \in \mathbb{R}^{m\times m}$, $\mathrm{Lip}(B) \leq m$. Hence $\Gamma(J_m) \leq m$.

For the reverse inequality, let $J_m = B \circ A$ be any multilinear Hilbert factorization, and set $a_{ij} = A(e_i, e_j)$. For independent Rademacher vectors $\varepsilon, \delta \in \{-1, 1\}^m$, orthogonality gives

$$\mathbb{E}\|A(\varepsilon, \delta)\|_2^2 = \sum_{i,j=1}^{m} \|a_{ij}\|_2^2 \leq m^2\|A\|^2.$$

Thus some sign pair satisfies $\|A(\varepsilon, \delta)\|_2 \leq m\|A\|$. As $A(0, \delta) = 0$ and $\|J_m(\varepsilon, \delta)\|_1 = m^2$,

$$m^2 \leq \mathrm{Lip}(B)\|A(\varepsilon, \delta)\|_2 \leq m\|A\|\,\mathrm{Lip}(B).$$

Taking the infimum gives $\Gamma(J_m) \geq m$. □

Corollary 5 (Negative answer to Question 1)**.** *The comparison $T \mapsto f_T$ is not an isometry. In fact, for $m = 2^{15} = 32768$,*

$$\gamma_2^{\mathrm{Lip}}(f_{J_m}) \leq 120\sqrt{2m} = 30720 < 32768 = \Gamma(J_m).$$

Moreover, the ratio $\Gamma(J_m)/\gamma_2^{\mathrm{Lip}}(f_{J_m})$ is unbounded as $m \to \infty$.

6 A single operator outside the multilinear ideal

Choose any sequence $m_k \to \infty$ and put

$$c_k = 120\sqrt{2m_k}, \qquad \omega_k = c_k^{-1}.$$

Let

$$X = \left(\bigoplus_{k=1}^{\infty} E_{m_k} \right)_{\ell_1}, \qquad Z = \left(\bigoplus_{k=1}^{\infty} \ell_1^{m_k \times m_k} \right)_{\ell_2}.$$

For $x = (x_k)$ and $y = (y_k)$ in X, define

$$T(x,y) = \big(\omega_k x_k \otimes y_k\big)_{k=1}^{\infty} \in Z. \tag{2}$$

This is bounded because

$$\|T(x,y)\|_Z \le \sup_k \omega_k \left(\sum_k \|x_k\|_1^2 \|y_k\|_1^2 \right)^{1/2} \le \sup_k \omega_k \, \|x\|_X \|y\|_X.$$

Theorem 6 (Negative answer to Question 2)**.** *For the bilinear operator T in (2), its associated Segre operator f_T factors metrically through a Hilbert space, with $\gamma_2^{\mathrm{Lip}}(f_T) \le 1$, but*

$$T \notin \Gamma(X, X; Z).$$

Proof. Normalize Proposition 3 by setting $G_m = F_m/(24\sqrt{2})$. Then

$$\mathrm{Lip}(G_m) \le 1, \qquad \mathrm{Lip}(G_m^{-1}) \le 120\sqrt{2m}. \tag{3}$$

For $p = x \otimes y \in \Sigma_{X,X}^{\pi}$, let $p_k = x_k \otimes y_k \in \Sigma_{m_k}$ be its kth diagonal block and define

$$\mathcal{A}(p) = \big(G_{m_k}(p_k)\big)_k \quad \text{in} \quad \mathcal{H} = \left(\bigoplus_k H_{m_k} \right)_{\ell_2}.$$

The diagonal coordinate projection from $X \widehat{\otimes}_\pi X$ onto $\big(\bigoplus_k \ell_1^{m_k \times m_k}\big)_{\ell_1}$ is contractive. Thus, for $p, q \in \Sigma_{X,X}^{\pi}$,

$$\begin{aligned} \|\mathcal{A}(p) - \mathcal{A}(q)\|_{\mathcal{H}} &\le \left(\sum_k \|p_k - q_k\|_1^2 \right)^{1/2} \\ &\le \sum_k \|p_k - q_k\|_1 \le \pi(p - q). \end{aligned}$$

Hence $\mathrm{Lip}(\mathcal{A}) \le 1$.

Define $\mathcal{B}$ on $\mathcal{A}(\Sigma_{X,X}^{\pi})$ by

$$\mathcal{B}(\mathcal{A}(p)) = (\omega_k p_k)_k.$$

This is well-defined because every G_{m_k} is injective. By (3),

$$\omega_k \|p_k - q_k\|_1 \le \|G_{m_k}(p_k) - G_{m_k}(q_k)\|_2.$$

Taking the ℓ_2 sum shows $\mathrm{Lip}(\mathcal{B}) \le 1$. Since $f_T = \mathcal{B} \circ \mathcal{A}$, we have $\gamma_2^{\mathrm{Lip}}(f_T) \le 1$.

Suppose that $T \in \Gamma(X, X; Z)$. Let $I_k : E_{m_k} \to X$ be the canonical isometric inclusion and $Q_k : Z \to \ell_1^{m_k \times m_k}$ the coordinate projection. The multi-ideal property gives

$$\Gamma\big(Q_k T(I_k, I_k)\big) \le \Gamma(T).$$

But

$$Q_k T(I_k x, I_k y) = \omega_k J_{m_k}(x, y),$$

so Proposition 4 yields

$$\frac{m_k}{120\sqrt{2m_k}} = \omega_k \Gamma(J_{m_k}) \le \Gamma(T).$$

The left side tends to infinity, a contradiction. □

7 Verification notes and limitations

- The proof is self-contained apart from the definitions and multi-ideal property already established in the source paper. The key points for review are Lemma 1, the quadratic identity in Lemma 2, and the diagonal-block Lipschitz estimate in Theorem 6.
- The included script `code/check_rank_one_factorization.py` checks the displayed direction and radial inequalities on random real rank-one tensors and checks the exact finite-dimensional constants used in Corollary 5. These tests are sanity checks, not part of the proof.
- The construction is stated over $\mathbb{R}$. This suffices to disprove the unqualified real-or-complex questions as universal assertions. A complex analogue should replace the simultaneous sign quotient by a phase quotient, but that extension is not claimed here.

8 Novelty check

A bounded search was performed on 9 August 2026. The run indexes were searched for arXiv:1805.09748, "metric Hilbert factorization", "Segre cone", and the exact question phrases; no prior packet or attempt was found. Web/arXiv searches used the title, the exact Question 2 wording, the notation Γ_2^{Lip}, and combinations of "multilinear", "metric", "Hilbert", and "isometry". The OpenAlex citation record for the source listed four citing works: arXiv:1805.02115, arXiv:1808.04842, arXiv:1812.00114, and arXiv:2311.08800. Their titles/abstracts and searchable arXiv records concern Lipschitz summing operators, dominated operators, tensor-norm duality, and Bloch mappings; none states or supplies an answer to Questions 1–2. No later paper explicitly answering either question was found within these bounds. Novelty is therefore plausible, not certified.

9 Human-review recommendation

Review as a claimed full counterexample to both source questions. The most important audit is the passage from the quotient direction metric $\delta(a, b)$ to the Hilbert feature $h_m(a)$, followed by the assertion that the global map $\mathcal{B}$ is well-defined and 1-Lipschitz on the image of $\mathcal{A}$. If those steps pass, the Rademacher lower bound and block contradiction are elementary and the conclusion is complete.

References

[1] M. Fernández-Unzueta and S. García-Hernández, *Multilinear Operators Factoring through Hilbert Spaces*, arXiv:1805.09748; Banach J. Math. Anal. 13 (2019), 234–254.